

Xander Byrne

Innovative solutions to important challenges

Astrophysics PhD Student
University of Cambridge

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Education

University of Cambridge

Cambridge, 2019-23

- MSci Natural Sciences, 1st class with distinction
- Ranked top of the class in Astrophysics examinations
- Awarded *Holgate-Pollard Memorial Prize* from the University
- Awarded *Wilson Prize and Skerne (1745) Scholarship* from St Catharine's College

King Edward VI Camp Hill School for Boys

Birmingham 2012-19

- A-levels (all A*): Maths, Further Maths, Physics, Chemistry
- Twelve GCSEs, all A*/9

Experience

CDT in Data Intensive Science

Cambridge, 2023-24

- Extensive training in *PyTorch* and modern machine learning techniques
- Courses in data handling & pre-processing, (un)supervised learning, Bayesian modelling
- Applied state-of-the-art analysis techniques: MCMC, nested sampling, GPs
- Trained diffusion models for image generation, optimising architectures
- Small group teaching from leading experts in ML and statistics

Max Planck Institut für Astronomie – Summer Internship

Heidelberg, 2022

- First ever systematic discovery of a lensed high-redshift quasar
- Developed software based on contrastive learning, to isolate rare objects from a large dataset
- Successfully implemented cutting-edge ML techniques in *TensorFlow*
- Customised methods to specific use case, optimising architecture
- Explored various unsupervised ML techniques (e.g. VAEs, SOMs)
- Employed a range of clustering algorithms (e.g. GMMs, DBSCAN)
- Participated in intensive workshops on Git and Bash
- Presented work at MPA's seminar series and two further seminars in Cambridge
- Published in *Monthly Notices of the Royal Astronomical Society* (Byrne et al., 2024)

Other Projects

Masters' Project / Summer Internship

Cambridge, 2023

- Secured over £3k of funding to continue masters' project on Venus-like exoplanets
- Built Python wrappers for Fortran code; simulating surface-atmosphere interactions

International Chemistry Olympiad

Paris, 2019

- Represented the UK in the nation's four-person team IChO 2019
- Assimilated a huge amount of knowledge in a short timeframe
- Ranked 41st in the world; awarded silver medal from the Monnaie de Paris